Name: _____

TOPIC TEST 1

Chapter 5 Binomial distribution

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (29 marks).
- Part C: Two analysis questions (16 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 In a test of ESP (extrasensory perception), an experimenter looks at cards that are hidden from the subject. Each card has a star, a triangle, a circle or a square printed on it. The experimenter looks at 10 of these randomly selected cards in turn and the subject tries to read the mind of the experimenter and name the shape on each card. What is the probability that the subject gets more than three correct if the subject does not have ESP and is just guessing?

A 0.2505

B 0.2241

C 0.3150

D less than 0.0001

E 0.4744

[1 mark]

2 A fair six-sided die is rolled 18 times. The number of times a multiple of 3 occurs has:

A a binomial distribution with $\mu = 8$

B a geometric distribution with $p = \frac{1}{3}$

C a hypergeometric distribution with $k = 4$

D a binomial distribution with $\sigma = 2$

E none of the above.

[1 mark]

3 Which of the following random variables is geometric?

A The number of phone calls received in a two-hour period

B The number of times a coin is tossed before two successive heads occur

C The number of cards draw from a standard deck of playing cards with replacement until a draw is spade

D The number that occurs when a six-sided die is rolled

E The number of cards dealt from a standard deck of playing cards so that at least one of the cards is a spade.

[1 mark]

4 A six-sided die is rolled four times. The probability that a number less than 5 occurs on two occasions is:

A $\frac{2}{3}$

B $\frac{15}{81}$

C $\frac{4}{27}$

D $\frac{8}{27}$

E $\frac{4}{9}$

[1 mark]

5 A deck of 10 cards consists of five hearts and five spades. The cards are shuffled and the cards are progressively turned over one at a time beginning with the top card. Let X be the number of cards that are turned over until the first heart appears. The probability that X is greater than 6 is closest to:

A 0.5000

B 0

C 0.0040

D 0.00315

E 0.0155

[1 mark]

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Part B

Short-answer questions

Six short-answer questions
29 marks
Show your working where appropriate.

1 A researcher goes to a small school with 400 students. It is known that 25 of the students are carriers of a gene for a particular medical condition called Syndrome Z. The researcher selects a random sample of 50 students.

a Confirm that this is an example of a hypergeometric probability experiment.

b Calculate the values of N, n, X and k for this experiment.

c Calculate the probability that there are 7 students who are carriers of the gene for Syndrome Z in the sample.

[5 marks]

2 A golfer knows that, on average, she hooks her drive three in every 10 attempts. If X represents the number of hooked drives before she hits a straight one, find $P(2 \leq X \leq 5)$.

[5 marks]

3 The probability that a person currently in an Australian high security prison for committing a serious crime will reoffend within three years of release is known to be 0.68. Ten prisoners are randomly selected from a high security prison. Find the probability that at least four will reoffend within three years of their release.

[4 marks]

4 A binomial variable, X, has the probability function:

$P(X = x) = \binom{6}{x}(0.45)^x(0.55)^{6-x}$.

Find:

a the number of trials

b the probability of success in any trial

c the probability distribution as a table.

[4 marks]

5 A binomial experiment has 8 trials. The probability of success in any trial is 0.6. If X = number of successes, calculate the probability that:

a $X = 4$

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b X is at least 2

c X is no more than 2.

[7 marks]

6 Find the required probabilities in each of the following situations.

a A binomial experiment has 15 trials and $p = 0.25$. Find the probability that there are 4 successes obtained from the 15 trials.

b Find the binomial probability of 7 successes from 11 trials, where the probability of success is 0.85.

[4 marks]

Part C

Analysis questions

Two analysis questions
16 marks
Show your working where appropriate.

1 The Big Deal Corporation (BDC) does the majority of its business with Indonesia. There are two outcomes for inflation in Indonesia – high ($> 8\%$) and low ($\leq 8\%$). From years of experience, the owner of BDC knows that the company makes $50 million in profit for each year of high inflation and loses $20 million in each year of low inflation.

Economists estimate that the probability of high inflation in Indonesia is 80% for the next four years.

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a What is the probability that at least 3 of the next four years will have high inflation?

b What net profit can the owner expect the company to make in the next four years if there is high inflation?

[8 marks]

2 A keen archer knows that she scores a bullseye one in every four shots.

a If she has 7 shots at the target, what is the probability she hits the bullseye at least twice?

b How many shots will she need to take in order to ensure a probability of more than 0.9 of scoring at least one bullseye?

[8 marks]

[Total marks: 50]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 This is a binomial experiment with $n = 10$ and $p = 0.25$.

Let X be the number of correct guesses.

$$\begin{aligned} \text{P}(X > 3) &= 1 - P(X \leq 3) \\ &= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)] \\ &= 1 - [0.0563\ldots + 0.1877\ldots + 0.2815\ldots + 0.2502\ldots] \\ &\approx 0.2241 \end{aligned}$$

$\therefore$ **B** **[1 mark]**

2 This is a binomial experiment with $n = 18$ and $p = \frac{1}{3}$.

Let X be the number of correct guesses.

$$\mu = np = 6$$
$$p = \frac{1}{3}$$
$$\sigma = \sqrt{npq} = \sqrt{18 \times \frac{1}{3} \times \frac{2}{3}} = 2$$

$\therefore$ **D** **[1 mark]**

3 The variable (X) in a geometric probability experiment is the number of failures before the first success (or the number of trials for the first success to occur).

p and q must be constant.

$x = 0, 1, 2, 3, \ldots, n$

These conditions are satisfied by the situation in C.

$\therefore$ **C** **[1 mark]**

4 This is a binomial experiment with $p = \frac{2}{3}$, $q = \frac{1}{3}$, $n = 4$ and $x = 2$.

$$P(X = 2) = \binom{4}{2}\left(\frac{2}{3}\right)^2\left(\frac{1}{3}\right)^2 = \frac{8}{27}$$

$\therefore$ **D** **[1 mark]**

5 There are only 5 spades, so the greatest value that X can be is 6.

$\text{P}(X > 6) = 0$

$\therefore$ **B** **[1 mark]**

Part B – Short-answer questions

1 a In this experiment there is:

- a finite population of 400 to be sampled
- for each trial there are only two possible outcomes – students who are gene carriers and those who are not
- there are exactly 25 successes in the population
- a sample size of 50 is obtained from the population without replacement.

This is a hypergeometric experiment. **[1 mark]**

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b N = population = 400

n = sample size = 50

X has a maximum value of 25, so $X = 0, 1, 3, \ldots, 25$

k = number of successes in the population = 25 **[1 mark]**

c $P(X = x) = \dfrac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$ **[1 mark]**

$$P(X = 7) = \frac{\binom{25}{7}\binom{400-25}{50-7}}{\binom{400}{50}}$$

$$= \frac{\binom{25}{7}\binom{375}{43}}{\binom{400}{50}}$$ **[1 mark]**

≈ 0.0184 **[1 mark]**

2 This is a geometric experiment.

Let X = number of hooks before a straight drive. **[1 mark]**

$p = 0.7$ and $q = 0.3$ **[1 mark]**

$P(2 \leq X \leq 5) = P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5)$ **[1 mark]**

$= (0.3)^2(0.7) + (0.3)^3(0.7) + (0.3)^4(0.7) + (0.3)^5(0.7)$

≈ 0.0893 **[1 mark]**

3 Let X be the number of prisoners who reoffend.

$n = 10$

$p = 0.68$ **[1 mark]**

$P(X = x) = \binom{10}{x}(0.68)^x(0.32)^{10-x}$ **[1 mark]**

$P(X \geq 4) = 1 - P(X < 4)$

$= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)]$ **[1 mark]**

$$= 1 - \left[\binom{10}{0}(0.32)^{10} + \binom{10}{1}(0.68)^1(0.32)^9 + \binom{10}{2}(0.68)^2(0.32)^8 + \binom{10}{1}(0.68)^3(0.32)^7\right]$$

[1 mark]

$\approx 1 - (0.000\,011 + 0.000\,239 + 0.002\,288 + 0.012\,965)$

$= 0.9845$ **[1 mark]**

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4 a For any binomial experiment $P(X = x) = \binom{n}{x} p\, q^{n-x}$

For this binomial experiment $P(X = x) = \binom{6}{x} (0.45)^x (0.55)^{6-x}$

$n = 6$ **[1 mark]**

b $p = 0.45$ **[1 mark]**

c $P(X = 0) = \binom{6}{0} (0.45)^0 (0.55)^6 \approx 0.0277$

$P(X = 1) = \binom{6}{1} (0.45)^1 (0.55)^5 \approx 0.1359$

$P(X = 2) = \binom{6}{2} (0.45)^2 (0.55)^4 \approx 0.2780$

And so on. **[1 mark]**

x	0	1	2	3	4	5	6
$p(x)$	0.0277	0.1359	0.2780	0.3032	0.1861	0.0609	0.0083

[1 mark]

5 a This is a binomial experiment.

$n = 8$

$p = 0.6$

$q = 0.4$ **[1 mark]**

$P(X = x) = \binom{n}{x} p^x q^{n-x}$

$P(X = 4) = \binom{8}{4} (0.6)^4 (0.4)^4$ **[1 mark]**

$= 0.232\,243\ldots$

≈ 0.2322 **[1 mark]**

b $P(X \text{ is at least } 2) = P(X \geq 2)$

$= 1 - P(X < 2)$

$= 1 - [P(X = 0) + P(X = 1)]$ **[1 mark]**

$P(X = 0) = \binom{8}{0} (0.6)^0 (0.4)^8 = 0.000\,655\ldots$

$P(X = 1) = \binom{8}{1} (0.6)^1 (0.4)^7 = 0.007\,864\ldots$

$P(X \geq 2) = 1 - (0.000\,655\ldots + 0.007\,864\ldots)$

≈ 0.9915 **[1 mark]**

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c $P(X \text{ is no more than } 2) = P(X \leq 2)$ **[1 mark]**

$P(X = 2) = \binom{8}{2}(0.6)^2(0.4)^6 = 0.041\,288\ldots$

$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

$= 0.000\,655\ldots + 0.007\,864\ldots + 0.041\,288\ldots$

≈ 0.0498 **[1 mark]**

6 a This is a binomial experiment.

$p = 0.25$

$n = 15$

$x = 4$ **[1 mark]**

Intersection of row where $x = 4$ (from the left) and column where $p = 0.25$ (from above)

$= 0.2252$

$P(X = 4) \approx 0.2252$ **[1 mark]**

b This is a binomial experiment.

$p = 0.85$

$n = 11$

$x = 7$ **[1 mark]**

Intersection of row where $x = 7$ (from the right) and column where $p = 0.85$ (from below)

$= 0.0536$

$P(X = 7) \approx 0.0536$ **[1 mark]**

Part C – Analysis questions

1 a This situation is a binomial experiment.

$n = 4$

$p = 0.8$ and $q = 0.2$ **[1 mark]**

$P(X \geq 3) = P(X = 3) + P(X = 4)$ **[1 mark]**

$= \binom{4}{3}(0.8)^3(0.2) + (0.8)^4$

$= 0.4096 + 0.4096$

$= 0.8192$ **[1 mark]**

b $E(X) = np$ **[1 mark]**

$= 4 \times 0.8$

$= 3.2$ **[1 mark]**

X can only have the values 0, 1, 2, 3 or 4.

$E(X) = 3$ **[1 mark]**

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There are 3 years of high inflation and 1 year of low inflation.

Profit $= 3 \times \$50$ million $- 1 \times \$20$ million **[1 mark]**

$= \$130$ million

The company can be expected to generate a net profit of \$130 million in the next four years. **[1 mark]**

2 a This is an example of a binomial experiment.

$n = 7$

$p = 0.25, q = 0.75$

$X =$ number of bullseyes **[1 mark]**

$P(\text{at least 2 bullseyes}) = P(X \geq 2)$

$= 1 - P(X < 2)$ **[1 mark]**

$= 1 - [P(X = 1) + P(X = 0)]$

$= 1 - \binom{7}{1}(0.25)^1(0.75)^6 - (0.75)^7$

$= 1 - 0.3134\ldots - 0.1314\ldots$

≈ 0.5551 **[1 mark]**

b $n = ?$

$p = 0.25, q = 0.75$

$X =$ number of bullseyes **[1 mark]**

$P(X \geq 1) > 0.9$

$P(X = 1) + P(X = 2) + P(X = 3) + \ldots > 0.9$

$1 - P(X = 0) > 0.9$ **[1 mark]**

$1 - (0.75)^n > 0.9$

$-(0.75)^n > 0.9 - 1$

$(0.75)^n < -0.9 + 1$

$(0.75)^n < 0.1$ **[1 mark]**

$n > 8.0039\ldots$ (using a graphics calculator or trial and error) **[1 mark]**

The archer must take at least 9 shots to ensure the probability of scoring at least one bullseye is at least 0.9. **[1 mark]**

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